Post-Quantum Mathematics

Reading Guide

Compiled by the Spaced Research Ledger

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Post-Quantum Mathematics covers the layer underneath the migration. A companion collection, Post-Quantum Cryptography Migration, covers the deployment of the new algorithms across real systems. This one covers where those algorithms come from, who is choosing them, what the alternatives are, and what is being broken.

The distinction is practical rather than academic. Anyone deploying the new cryptography needs to know that ML-KEM is the standard.

Anyone deciding what to deploy for the next twenty years needs more. Two European governments recommend an algorithm NIST rejected. A fifth NIST algorithm was selected but never written up. One candidate signature scheme was withdrawn in July 2026, after an AI-assisted attack halved its security margin.

Three things run through the whole collection.

The first is concentration risk. The two algorithms the world is deploying both rest on the difficulty of problems over structured lattices. Every alternative in Part 3, and much of the policy in Part 2, exists because that is a single point of failure.

The second is that standardization is plural. NIST is the most visible body, not the only one. South Korea picked its own winners. China is running its own process, with a stated intent to shape norms rather than follow them. ISO has standardized two schemes NIST declined.

The third is that breaking things is the process working. Part 5 reads like a list of failures. It is a list of candidates being tested before anyone depends on them, which is exactly what a standardization process is for.

Every statement in every part carries a numbered citation to a public source, listed at the end of that part. When sources disagree, the disagreement is reported under its own heading rather than resolved by guesswork. Each part states when it was last updated.

How to Read This Collection

Each part opens with a summary written for someone who reads nothing else. Numbered sections follow, and each begins with a short italic paragraph explaining in plain language what the thing is and why it matters. Inside each section, Recent Developments covers what changed lately and Current State covers what is durably true. Unresolved Conflicts, where present, records disagreements between sources with a note on what would settle each one.

Technical terms are explained once, on first appearance, and then used normally. Security levels and attack costs are given in the notation the sources used, because in cryptography the exponent is the finding.

Start with Part 1 if you want to know what is settled. Start with Part 5 if you want to know what is not.

The Parts

Standards Tracks and Candidates

Part 1 - NIST Standards Track. The three ratified standards and the errata now being applied to two of them. Also the signature standard still in review, the fifth algorithm selected but unpublished, and the nine candidates still competing. Also the IETF work that turns an algorithm into something a protocol can call.

Part 2 - Non-NIST National & Regional Standards Tracks. South Korea's own competition and its move into silicon, and China's second process. Also Germany's harder deadlines, France's decision to tie product certification to quantum resistance, and Japan's dual track. The algorithms are converging. The requirements around them are not.

Part 3 - Candidate Asymmetric Algorithms. The schemes NIST did not choose, two of which ISO standardized anyway. Covers the code-based family, with its 45-year unbroken record and its newly narrowed margin. Also the isogeny family that survived its own most famous break, and the conservative lattice alternatives two European governments prefer.

Primitives and Cryptanalysis

Part 4 - Symmetric & Hash-Based Research. Why the reassuring summary about symmetric cryptography is right about AES and wrong about many of the modes built on it. Covers Grover's quadratic speedup and the exponential speedups that apply to specific constructions under a strong attacker model. Also which constructions now carry quantum-era security proofs, and the hash functions the ratified standards use internally.

Part 5 - Cryptanalysis of Candidate & Non-Ratified Schemes. What is being attacked and how far the attacks have got. Includes the one withdrawal in this period, and the sustained pressure on the multivariate family that makes up four of the nine remaining candidates. Also two August 2026 results narrowing the margin on the oldest scheme here.

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from three to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.